

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-  
a50e92db5df29be90.jsonl`  
- SHA-256: `b4c4cbe8c8ab54fcb0e93259672deea6cc817b7247e0797a6e46e234e1dcd4d`  
- Source modified: `2026-06-06T17:32:18+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:31:08.847Z)

Source Extractor

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Fetch and extract key claims from this source:

****URL:**** https://www.irisa.fr/metiss/gravier/biblio/03/kijak-icme03.pdf

****Title:**** HMM-Based Structuring of Tennis Videos Using Visual and Audio Cues (Kijak et al., ICME)

****Found via:**** practitioner/local-levers search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:11.409Z)

I'll fetch the source and extract key claims relevant to the research question.

3. user (2026-06-06T17:31:25.402Z)

I appreciate your detailed request, but I need to be direct: the content you've provided is a corrupted or improperly encoded PDF file. The text is displayed as compressed binary data and is not readable in its current form.

What I can see:

- The document appears to be a multi-page PDF (pages 6, 29, 56, 73 referenced)
- There are compressed stream objects and PDF structural elements
- The actual text content is encoded/compressed and cannot be parsed for the specific claims you're seeking

****To help you extract this information, I would need:****

1. A properly decoded/extracted text version of the paper, or
2. The PDF file in a format that can be properly rendered, or
3. The paper's title/authors/DOI so I can help you locate a readable version

****What I cannot do:****

- Decode the binary PDF compression without proper decompression tools
- Extract specific quotes about HMM methods, audio cues, visual cues, accuracy metrics, or datasets from unreadable source material

If you can provide the paper title, authors, or a clean text extract, I'd be happy to identify those seven specific findings with proper citations and your requested character limits on quotes.

[Binary content (application/pdf, 188.7KB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767080102-6svilu.pdf]

4. assistant (2026-06-06T17:31:28.275Z)

The WebFetch couldn't parse the PDF, but it was saved locally. Let me extract the text directly from the downloaded file.

5. user (2026-06-06T17:31:31.865Z)

3.13.13 (tags/v3.13.13:01104ce, Apr 7 2026, 19:25:48) [MSC v.1944 64 bit (AMD64)]

WARNING: Package(s) not found: pdfplumber
WARNING: Package(s) not found: PyPDF2
WARNING: Package(s) not found: pymupdf

6. assistant (2026-06-06T17:31:34.603Z)

No PDF libraries installed. Let me try installing pymupdf, which is the most reliable for text extraction.

7. user (2026-06-06T17:31:38.895Z)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

8. user (2026-06-06T17:31:42.958Z)

PAGES: 4

===== PAGE 1 =====

HMM BASED STRUCTURING OF TENNIS VIDEOS USING VISUAL AND AUDIO CUES

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ABSTRACT

This paper focuses on the use of Hidden Markov Models (HMMs) for structure analysis of videos, and demonstrates how they can be efficiently applied to merge audio and visual cues. Our approach is validated in the particular domain of tennis videos. The basic temporal unit is the video shot. Visual features are used to characterize the type of shot view. Audio features describe the audio events within a video shot. The video structure parsing relies on the analysis of the temporal interleaving of video shots, with respect to prior information about tennis content and editing rules. As a result, typical tennis scenes are identified. In addition, each shot is assigned to a level in the hierarchy described in terms of point, game and set.

1. INTRODUCTION

Video structure parsing consists in extracting logical story units from the considered video. It's a mandatory step to efficiently organize and retrieve video contents. The structure to be estimated relies on the nature of the video. In this paper, we address the problem of recovering sport video structure that can be useful in a further indexing task. Sport video analysis is motivated by the broadcasters needs of a detailed annotation of video contents. This annotation is usually used to select relevant excerpts in order to easily build summaries or magazines. Up to now, this logging task is performed manually.

Broadcasted video data consist of several multimodal information streams.

Numerous approaches presented in the literature have discussed how to extract high-level semantic events from video. To achieve such a goal, algorithms have to be dedicated to one particular type of videos such as sports or news. In this context, most of the approaches use the individual visual or audio cues. Only a few attempts dealing with multimodal analysis of audio-visual information have appeared recently, such as [1] on sports,

or [2] on news. These approaches rely on a successive use of visual and audio features. In a first step, visual features are used to detect plays in the original row. In a second step, a detection of audience or commentator excitement is used to select the most exciting plays [3]. The reverse scheme has also been proposed in [4]: cheering sounds are first extracted to locate an important event. Visual analysis is then performed in cheering sounds regions to recognize the event.

In this paper, we propose a tennis scene classification and segmentation method that automatically labels each segment of a tennis video with one of the following four types: first missed serve, rally, replay, and break. Hidden Markov Models (HMMs) are used to merge audio-visual informations, and to represent the hierarchical structure of a tennis match. Domain knowledge is implicitly taken into account through the learning process associated with HMMs. This implicit approach is more suitable than heuristic rules oriented schemes because the learning process can derive more complete knowledge. In addition, previous works that use HMMs for video parsing or segmentation suggest good potential, even if only visual features are considered in sport domain [5, 6].

Our solution differs from existing methods in the following main points: (1) it combines visual/audio cues simultaneously; (2) different levels of classifications are generated over the entire tennis video at different levels; and (3), the segmented content are semantically meaningful to users.

This paper is organized as follows. Section 2 gives an overview of the system and briefly describes the exploited audio and visual features. Section 3 introduces the structure analysis mechanism. Experimental results are presented and discussed in section 4. Section 5 provides the concluding remarks.

2. SYSTEM OVERVIEW

In this section, we briefly describe the extraction of audio and visual cues. First, the video stream is automatically segmented into shots by detecting cuts and dissolve transitions. For each shot, we extract from the image stream, one

===== PAGE 2 =====

Video program

frames

Shot Detection

Shot Features

Extraction

HMM Training

HMM Parsing

Observations

Program Syntax

States

Transitions

Domain Knowledge

Fig. 1. Structure analysis system overview

keyframe along with its image features (color and motion intensity), and from the audio stream, sound classes such as ball hits, applause, or speech. The segmented video results in an observation sequence which is parsed by a HMM process (Figure 1). As a result, each shot is assigned to a level in the hierarchy described in terms of point, game and set, and is labelled with one of these four types: first missed serve, rally, replay, and break.

Such a tennis scene classification and segmentation is possible because tennis video is characterized by a typical production style, known as the tennis video syntax.

2.1. Tennis Syntax

Sport video production is characterized by the use of a limited number of cameras at almost fixed positions. Considering a given instant, the point of view giving the most relevant information is selected by the producer, and broadcast. Therefore sports are composed of a restricted number of typical scenes producing a repetitive pattern. For example, during a rally in a tennis video, the content provided by the camera filming the whole court is selected (we name it global view in the following). After the end of the rally, the player who has just carried out an action of interest is captured with a close-up. As close-up views never appear during a rally but right after or before it, global views are generally significant of a rally. Another example consists in replays that are notified to the viewers by inserting special transitions.

We identify here four typical pattern in tennis videos that are: first missed serve, rally, replay, and break. The break class includes the scenes unrelated to games, such as commercials. A break is characterized by an important succession of such scenes. It appears when players change ends, generally every two games. We also take advantage of the well-defined and strong structure of tennis broadcast. On the contrary of time-constrained sports that have a relatively loose structure, tennis is a score-constrained sport that can be broken down into sets, games and points (Figure 2).

The different types of views present in a tennis video can be divided into four principal classes: global, medium, Fig. 2. Structure of Tennis Game

close-up, and audience. In a tennis video production, global views contain much of the pertinent information. The remaining information relies on the presence or the absence of non global views but is independent of the type of these views.

2.2. Visual Features

We used visual features to identify the global views within all the extracted keyframes. The process can be divided into two steps. First, a keyframe K_{ref} representative of a global view is selected without making any assumption about the playing area color. Once K_{ref} has been found, each keyframe is characterized by a similarity distance to K_{ref} . In the following, we motivate the choice of the visual features we use, before briefly describing the identification process. Global views are characterized by a rather homogeneous color content (the colors of the court and its surrounding), although medium and audience views are characterized by scattered color content.

In addition, a global view must capture at each time the main part of the court, whereas in close-up views, the camera is generally tracking the player. The first class can thus be characterized by a small camera motion while the second implies important camera translations. Based on these observations, we choose two features to identify global views: (1) a vector of dominant colors F and its spatial coherency C , and (2) activity A that reflects camera motion during a shot.

The visual similarity measure $v(K1, K2)$ between two keyframes $K1$ and $K2$ is defined as a weighted function of the spatial coherency, the distance function between the dominant color vectors, and the activity:

$$v(K1, K2) =$$

(1)

$$w1 |C1 - C2| + w1 d(F1, F2) + w3 |A1 - A2|$$

where $w1$, $w2$, and $w3$ are the weights.

===== PAGE 3 =====

To find a keyframe K_{ref} representative of a global view, we consider dominant colors ratios. In a reduced subset of keyframes whose highest percentage of dominant color is more than 50%, we apply the least median square method, to minimize the median distance of all the remaining keyframes to a randomly selected keyframe. This leads to obtain K_{ref} (see details in [7]). The visual similarity measure $v(K_t, K_{ref})$, denoted by v_t in the following, is then computed between each keyframe K_t and K_{ref} .

2.3. Audio Features

Generally speaking, there are two approaches to combine audio and visual features for structure analysis: one is to combine audio and visual features into a single audiovisual feature vector before the classification. The other consists in classifying separately according to each modality before integrating the classification results.

In this work, an intermediate strategy is used which consists in extracting separately high level audio and visual cues, the classification being made using the audio and visual cues simultaneously. As mentioned previously, the video stream is segmented in a sequence of shots.

Since shot

boundaries are more suitable for structure analysis than boundaries extracted from the soundtrack, the shot is considered as the base entity and features describing the audio content for each shot are used to provide additional information. For each shot, a binary vector describing which audio classes, among speech, applause, ball hits, noise and music, are present in the shot is extracted from an automatic segmentation of the audio stream. The automatic soundtrack segmentation is carried out using a Viterbi based system with Gaussian mixtures for each sound class and combination of sound classes. The soundtrack is segmented independently and the vectors are created from this independent segmentation.

3. STRUCTURE ANALYSIS

Prior information is integrated by deriving syntactical basic elements from the tennis video syntax. We define four basic structural units: two of them are related to game phases (first missed serves and rallies), the two others denote non-game phases (breaks and replays). Each of these units is modelled by a HMM. These HMMs rely on the temporal relationships between shots.

Each HMM state models either a single shot or a dissolve transition between shots.

The observation consists

of three elements: the visual similarity vt between the shot keyframe and $Kref$, the shot duration dt , and the audio vector at which characterizes the presence or absence of the pre-determined audio events. More formally, the probability of an observation ot conditionally to state j is then given by $b_j(ot) = p(vt|j) p(dt|j) P[at|j]$

(2)

MATCH

S

Br

S

Br

S

G

Br

G

G

Br

G

G

Br

G

G

P

P

P

P

P

SET

BREAK

First

serve

and rally

Replay

Rally

GAME

POINT

Fig. 3. Content hierarchy of broadcast tennis video where $p(vt|j)$ and $p(dt|j)$ are given by a smoothed histogram, and $P[at|j]$ is the product over each sound class k of the discrete probability $P[at(k)|j]$. Segmentation and classification of the whole observed sequence $o = o_1o_2...o_T$ into the different structural elements are performed simultaneously using a Viterbi algorithm. The most likely state sequence $s = s_1...s_T$ is given by

$$s = \arg \max_s \ln p(s) + \sum_t \ln bst(o_t) \quad (3)$$

To take into account the long-term structure of a tennis game, the four HMMs are connected to a higher-level HMM which represents the tennis syntax as illustrated in Figure 3. The transition probabilities between states of the higher-level HMM result entirely from a priori information, while the transition probabilities of the sub-HMMs are estimated.

4. EXPERIMENTAL RESULTS

Experimental data are composed of about 2 hours of manually labelled tennis video. About half of the data is used to train the HMM while the remaining half is reserved for the tests.

Recall and precision rates are given in Table 1 for three experiments: visual features only, audio features only and combined audiovisual approach. Shot duration is considered for all experiments. Experiments using audio features are sub-divided into two parts: using audio features manually annotated (denote by ?man.? in Table 1) to validate the model, and using audio features extracting from automatic segmentation (denote by ?segm.? in Table 1). The audio probabilities are estimated from the manually annotated audio features.

===== PAGE 4 =====

Visual	
Audio features	
Audio-visual	
features	
man.	
segm.	
man.	
segm.	
Segmentation	
71%	
68%	
44%	
78%	
66%	
accuracy	
Classification	
precision	
recall	
precision	
recall	
precision	

recall
precision
recall
precision
recall
First serve
67%
63%
92%
66%
66%
5%
87%
75%
71%
41%
Rallies
97%
54%
95%
63%
62%
28%
96%
83%
82%
65%
Replay
100%
80%
68%
38%
75%
15%
81%
88%
86%
82%
Break
51%
93%
91%
81%
30%
90%
94%
81%
68%
92%

Table 1. Classification and segmentation results with visual features only, audio features only and both audio-visual features

We observed that one single stream cannot provide enough information to identify different scenes. Rallies high precision rate (97%) and breaks recall rate (93%) show that low level visual features are efficient to identify global views from others. However, these features are not sufficient to distinguish a rally from a simple view of the court or from a first serve, as indicated by rallies low recall rate (54%). This means also that many global views are missed using only visual features. Replays detection relies essentially on dissolve transitions identification. The high precision rate corresponds to correctly detected dissolve transitions. Classification using reliable audio features increases precision rates for first missed serve. Comparable results are obtained for rallies detection. This suggests that audio fea-

tures are effective to describe rally scenes. Indeed, a rally is essentially characterized by the presence of ball hits and applause which happen at the end of the exchange, although a missed 1st serve is only characterized by the presence of ball hits. Therefore, there are less confusion between missed 1st serves and rallies.

Breaks precision rate increases also. Since break is the only state characterized by the presence of music, the break state works in this case like a commercial detector, and avoids false detections. On the contrary, replays are not characterized by a representative audio content, and almost all replays are missed.

The approach using only audio features automatically segmented gives poor performance. Recall rate for audio segmentation as described in 2.3 is 76.9%, and precision is 46.3%. Ball hits and music are well classified, whereas noise is often classified as ball hits. The errors from the automatic audio segmentation spread over the structuration process and drag the performance down. Another source of errors is the mismatch between the manually segmented audio features on which the HMM parameters were estimated, and the automatically generated audio features. An improvement was obtained by estimating the HMM parameters using automatically generated audio features, but it did not outclass the video only system.

Nevertheless, integrating multiple cues from different media improves performance. In particular, ball hits detection improves classification rates for rallies and 1st serves. And music detection prevents from false detection of breaks.

5. CONCLUSION

In this paper, we have presented an audio-visual model for tennis video structure parsing. Tennis structure is modelled by an HMM that integrates a priori information about tennis content and editing rules. HMMs provide an efficient framework to merge audio and visual cues, and to perform simultaneously classification and segmentation into scenes. Experiments show that audio information improve the performance of video classification, when features are quite reliable. Indeed errors from feature extraction spread over the entire analysis process. It is therefore necessary to enhance audio segmentation and classification.

To extend our tennis video structure analysis, we also intend to include a more sophisticated audio-video fusion model by refining audio and video content analysis.

6. REFERENCES

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9. assistant (2026-06-06T17:31:50.821Z)

I have the full text of the paper. This is the primary research paper (ICME 2003, Thomson multimedia R&D + IRISA). Let me extract the falsifiable claims relevant to the research question.

Key findings relevant to the research question (audio-based hit detection, HMM segmentation, audio+visual fusion for rally segmentation):

1. **"Audio-only automatic segmentation gives poor performance"** ? recall 76.9%, precision 46.3%; noise often misclassified as ball hits. This directly bears on the "audio-based hit detection" lever in the research question.
2. **"HMM fusing audio+visual cues"** for segmenting tennis into first-serve/rally/replay/break, with the Viterbi-based decoding.
3. **"Audio features (ball hits + applause) effectively characterize rally scenes"** ? multimodal fusion improves rally classification.
4. **"Audio-visual combined segmentation accuracy reached 78% (manual audio) vs 71% visual-only"**; rallies precision/recall 82%/65% combined.
5. **"Errors from automatic audio feature extraction propagate and drag overall performance below the video-only system"** ? a key cautionary finding for an audio-first local pipeline.

10. user (2026-06-06T17:32:05.917Z)

Structured output provided successfully

11. assistant (2026-06-06T17:32:18.199Z)

I've extracted and returned the key claims from the Kijak et al. ICME 2003 paper via the StructuredOutput tool.

Summary of what I found and why it matters to the research question:

- **"Source quality: primary"** ? peer-reviewed conference paper (IEEE ICME 2003) from Thomson multimedia R&D + IRISA. The original WebFetch couldn't parse the binary PDF, so I extracted the full text locally with pymupdf from the downloaded file at `C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\tool-results\webfetch-1780767080102-6svilu.pdf`.
- **"Most decision-relevant finding (central):"** their fully-automatic audio-based ball-hit/event classifier scored only **"76.9% recall / 46.3% precision"**, with **"noise frequently misclassified as ball hits"**, and these errors propagated downstream so the audio-only system did **"not"** beat the visual-only system. This is a direct, falsifiable caution for the "audio-first hit detection" lever in your pipeline ? audio cues helped only when fused, not standalone.
- **"Complementary finding (central):"** ball hits + end-of-rally applause are genuinely discriminative for rally vs. non-rally, and HMM fusion of audio+visual improved rally classification ? supporting audio as a **"complementary"** signal rather than a primary one.
- **"Numbers worth noting:"** visual-only gave high rally **"precision (97%) but low recall (54%)"** ? the same precision/recall asymmetry you're seeing with WASB candidates; audio fusion recovered recall (combined rallies 82%/65%). Caveat: tiny eval base (~2 hours of tennis video, half train / half test), and this is a broadcast-TV approach leaning on editing conventions (camera-shot syntax, dissolves) that won't transfer to raw self-recorded amateur badminton.